

Syntax

- The **signature** $\mathcal{L}$ now consists of:
 - set of sorts, denoted by α, β
 - constants, denoted by a, b, c , each having a sort $c : \alpha$
 - function symbols, denoted by f, g , each having a type $\alpha_1 \times \dots \times \alpha_n \mapsto \alpha$
 - predicate symbols, denoted by p, q , each having a type $\alpha_1 \times \dots \times \alpha_n$
- **Quantifiers**:
 - Given a formula G , a variable x , then $\mathcal{Q}x : \alpha$ is a quantified formula with $\mathcal{Q} \in \{\forall, \exists\}$
 - A formula is called quantifier-free (or ground) if it contains no occurrences of quantifiers

Semantics

- The **interpretation** I maps:
 - Each sort α to a non-empty set $I(\alpha)$, called domain of the sort
 - Each constant $c : \alpha$ to $c' \in I(\alpha)$
 - Each variable $x : \alpha$ to $x' \in I(\alpha)$
 - Each function symbol $f : \alpha_1 \times \dots \times \alpha_n \mapsto \alpha$ to a function $f' : I(\alpha_1) \times \dots \times I(\alpha_n) \mapsto I(\alpha)$
 - Each predicate symbol $p : \alpha_1 \times \dots \times \alpha_n$ to a relation $p' \subseteq I(\alpha_1) \times \dots \times I(\alpha_n)$
- Given I and a term t of a sort α , define the element $t' \in I(\alpha)$ as follows:
 - For constants and variables $t' = I(t)$
 - $f(t_1, \dots, t_n)' = f'(t_1', \dots, t_n')$
- Given I and a formula G , define the boolean value as follows:
 - For an atomic formula $p(t_1, \dots, t_n)$, define $p(t_1, \dots, t_n)' = 1 = (t_1', \dots, t_n') \in p'$
 - $G = (Q_1 \odot Q_2)' = 1$ if $Q_1' \odot Q_2'$ is satisfied
- **Quantifiers**
 - Given a set sequence of variables $\overline{x}$, two interpretations of the same signature are $\overline{x}$ -variant if they coincide on all symbols and variables not $\overline{x}$.
 - $(\forall x A)' = 1$ if for all $\overline{x}$ -variants I' of I we have $(A)' = 1$
 - $(\exists x A)' = 1$ if for some $\overline{x}$ -variant I' of I we have $(A)' = 1$

Axiomatic Interpretations

Restrict interpretations to those which satisfy the following set of formulas (so called axioms):

- reflexivity : $x = x$
- symmetry : $x = y \rightarrow y = x$
- transitivity : $x = y \wedge y = z \rightarrow x = z$
- congruence axioms for each function symbol f :

$$\left(\bigwedge_{i=1}^n x_i = y_i \right) \rightarrow f(x_1, \dots, x_n) = f(y_1, \dots, y_n)$$

- congruence axioms for each predicate symbol p :

$$\left(\bigwedge_{i=1}^n x_i = y_i \wedge p(x_1, \dots, x_n) \right) \rightarrow p(y_1, \dots, y_n)$$

Interpretation Classes

Restrict interpolations to those which satisfy certain properties

- **Example** :
 - Given the formula $f(x) = f(x+1) \wedge f(a) \neq f(b)$
 - 1. We consider the class of interpretations I where I interprets $+$ and 1 in the standard way over the integers. The formula is satisfiable in:
 - $I(a) = 1, I(b) = 2, I(x) = 3$
 - $f(x) = \begin{cases} 3 & \text{if } x = 4 \\ x & \text{else} \end{cases}$
 - 2. We consider the class of interpretations of example 1. but where $f(x) = f(x+1)$ is valid in I . The formula is not satisfiable as $f(x) = f(x+1)$ would impose that f maps all variables to the same value, in which case $f(a) \neq f(b)$ can not hold.
- **Satisfiability and Interpretation - Examples**
 - Given the formula $y = x + 1$ over integers, where x, y are variables, $+$, 1 are interpreted in the standard way over integers and the interpretation $I(x) = I(y) = 0$. It then holds that:
 - $I \not\models x = y + 1$
 - $I \models x = 1$ is satisfiable in I under the $\overline{x}$ -variant: $I'(x) = 1, I(y) = 0$
 - $x = y + 1$ is not valid in I , since $I \not\models x = y + 1$
 - Given the formula $z \neq a \rightarrow z = b$, where $a : \alpha, b : \alpha$ are constants and $z : \alpha$ is a variable. This formula is valid under the class of interpretations I for which it holds that:
 - $D_a^I = \{0, 1\}, I(a) = 0, I(b) = 1$
 - Given the formula $y = a \rightarrow y = b$, where a, b are constants and y is a variable. This formula is valid under the class of interpretations I for which it holds that :
 - $I(a) = I(b)$

Theory

- A theory is defined as
 - a signature Σ_T
 - a set of formulas A_T over Σ_T , called axioms of T

Unless specified otherwise, theories are quantifier-free

- An interpretation in which A_T are valid is called a T -interpretation
- If a formula G is valid in every T -interpretation, then G is T -valid (denoted by $T \models G$)

Theory of equality T_E :

- $\Sigma_E = \{a, b, \dots, f, g, \dots, p, \dots\}$
- $A_E = \left\{ \begin{aligned} &x = x, x = y \rightarrow y = x, x = y \wedge y = z \rightarrow x = z, \\ &\bigwedge_{i=1}^n y_i = x_i \rightarrow f(x_1, \dots, x_n) = f(y_1, \dots, y_n), \\ &\bigwedge_{i=1}^n y_i = x_i \wedge p(x_1, \dots, x_n) \rightarrow p(y_1, \dots, y_n) \end{aligned} \right\}$

A binary relation R over a set S is called a congruence relation, if:

- R is reflexive, transitive, symmetric and
- $\bigwedge_{i=1}^n y_i R x_i \rightarrow f(x_1, \dots, x_n) R f(y_1, \dots, y_n)$

The congruence class of $t \in S$ under R is $[t]_R = \{t' \in S \mid t R t'\}$,

- for which holds:
 - $\bigcup_{[t]_R} [t]_R = S$
 - $[t_1]_R \neq [t_2]_R \rightarrow [t_1]_R \cap [t_2]_R = \emptyset$

The congruence closure R^c of R is the relation for which holds:

- $R \subseteq R^c$
- $\forall R' : R \subseteq R' \rightarrow (R^c = R' \vee R^c \subseteq R')$

Theory of arrays T_A :

- $\Sigma_A = \{read, write\}$
 - where $read$ is a binary function and $write$ is ternary function
- $A_A = A_E \cup \left\{ \begin{aligned} &x = y \rightarrow read(write(A, x, v), y) = v, \quad \text{r-o-w}_1 \\ &x \neq y \rightarrow read(write(A, x, v), y) = read(A, y) \quad \text{r-o-w}_2 \end{aligned} \right\}$
 - $read(A, x)$ denotes the value of array A at pos. x
 - $write(A, x, v)$ denotes the modified array A with value v at pos. x

Given a function F with $write$ -terms $read(write(A, x, v), y)$, $F[F']$ denotes the formula where the $write$ -term has been replaced with F'

Congruence Closure Algorithm for T_A

T_A -satisfiability of F is reduced to T_E -satisfiability:

- If F contains $write$ -terms $read(write(A, x, v), y)$, use r-o-w axioms:

- 1. $F_1 : x = y \wedge F[v]$
- 2. $F_2 : x \neq y \wedge F[read(A, y)]$
- If both are unsat, then F is unsat

- If F doesn't contains $write$ -terms:
 - Use fresh function symbol f_A for array A
 - Replace all $read(A, x)$ with $f_A(x)$

Return T_E satisfiability of modified formula

Combination of Theories

Given:

- Given a set of theories $T_1, \dots, T_n$ in disjoint $\Sigma_1, \dots, \Sigma_n$ and its axioms $\bigcup_{i=1}^n A_{T_i}$

- Quantifier-free conjunction F of literals in $\bigcup_{i=1}^n \Sigma_i$

Checking if F is satisfiable in a T :

- Transform $F : \bigwedge_{i=1}^n F_i$ with F_i being a T_i formula, where $F_1, \dots, F_n$ are linked by variables

Example:

Given T_A, T_E and $F : (y = x + 2) \wedge (f(read(write(a, x, 3), y - 2)) \neq f(y - x + 1))$, we split:
 $F : (y = x + 2) \wedge f(c_4) \neq f(c_1)$
Shared equalities are derived e.g
 $-y = x + 2 \wedge c_2 = y - 2 \implies c_2 = x$

Linear Arithmetic	Uninterpreted Functions
$y = x + 2$ $c_1 = y - x + 1$ $c_2 = y - 2$ $c_3 = 3$	$f(c_4) \neq f(c_1)$
Arrays $c_4 = read(write(a, x, c_3), c_2)$	Shared Equalities $c_2 = x$ $c_4 = c_3$ $c_1 = c_2$

Split Reasoning Algorithm

```
procedure SeparatingReasoning(F)
input: formula  $F \in T_1 \cup \dots \cup T_n$ 
output: formulas  $F_1 \in T_1, \dots, F_n \in T_n$  s.t.
 $F$  and  $F_1 \wedge \dots \wedge F_n$  are equisatisfiable
assumptions: theory signatures  $\Sigma_1, \dots, \Sigma_n$  are disjoint
parameters: function  $head(t)$  returning the root symbol of a term  $t$ 
begin
  repeat as long as possible
    if  $F \in \Sigma_i$  and  $head(t) \in \Sigma_i$  with  $i \neq j$ :
      rewrite  $F[t_1, \dots, t_n]$  into  $F[f(t_1, \dots, t_n)] \wedge c = t$ ,
      where  $c$  is a fresh new variable
    if  $p \in \Sigma_i$  and  $head(t) \in \Sigma_i$  with  $i \neq j$ :
      rewrite  $F[p(t_1, \dots, t_n)]$  into  $F[p(t_1, \dots, t_n)] \wedge c = t$ ,
      where  $c$  is a fresh new variable
    if  $head(s) \in \Sigma_i$  and  $head(t) \in \Sigma_i$  with  $i \neq j$ :
      rewrite  $F[s = t]$  into  $F[s = c] \wedge c = t$ ,
      where  $c$  is a fresh new variable
    and repeat
  return modified  $F$  as  $F_1 \wedge \dots \wedge F_n$ , with each  $F_i \in T_{i_0, \dots, i_n}$ 
```

Not all theories can be easily combined this way

Convex Theory

A formula $F \in \mathcal{T}$ as a conjunction of

T -literals is convex if:

$$-T \models \left(F \rightarrow \bigvee_{j=1}^k (u_j = v_j) \right) \implies T \models F \rightarrow u_j = v_j$$

for some $j \in \{1, \dots, k\}$

Example

T_A e.g. is not convex, T_E e.g. is convex

(Proof of T_A not being convex is on the next page)

Stably infinite theories

A theory T_i is called stably infinite:
- For every T satisfiable formula $F \in T$, there exists a T -interpretation I :

- $I \models F$
- I has a domain of infinite cardinality

T_E, T_A, T_Z are stably infinite

Nelson Open method

Given $T_1, \dots, T_n$ decidable, quantifier-free theories with disjoint signature and which are stably infinite the Nelson Open Method for non-convex theories finds new equalities $u = v$ for shared u, v . Pass equality to other theories.

```
procedure NelsonOpenConvex(F)
input: conjunctive formula  $F \in T_1 \cup \dots \cup T_n$ 
output: satisfiable or unsatisfiable
assumptions: Nelson-Open restrictions over  $T_i$  hold
assumptions:  $F_i$  are convex
begin
  1.  $\overline{F}_1 \wedge \dots \wedge \overline{F}_n := SeparatingReasoning(F)$ 
  2. Run  $T_i$ -solver on  $\overline{F}_i$ . If a solver returns unsatisfiable, return unsatisfiable
  3. If there are shared variables  $u, v$  s.t.  $F_i \rightarrow u = v$  then
      $\overline{F}_i := \overline{F}_i \wedge u = v$ 
     go to step 2
  return satisfiable
end

procedure NelsonOpenNonConvex(F)
input: conjunctive formula  $F \in T_1 \cup \dots \cup T_n$ 
output: satisfiable or unsatisfiable
assumptions: Nelson-Open restrictions over  $T_i$  hold
assumptions:  $F_i$  are non-convex
begin
  1.  $\overline{F}_1 \wedge \dots \wedge \overline{F}_n := SeparatingReasoning(F)$ 
  2. Run  $T_i$ -solver on  $\overline{F}_i$ . If a solver returns unsatisfiable, return unsatisfiable
  3. If there are shared variables  $u, v$  s.t.  $F_i \rightarrow u = v$  then
      $\overline{F}_i := \overline{F}_i \wedge u = v$ 
     go to step 2
  4. If  $\overline{F}_i \rightarrow \bigvee_k u_k = v_k$ , for shared  $u_k, v_k$ , then
     if NelsonOpenNonConvex( $\overline{F}_1 \wedge \dots \wedge \overline{F}_n \wedge u_k = v_k$ ) is satisfiable, for any  $k$ 
     then return satisfiable
     else return unsatisfiable
  return satisfiable
end
```

SMT